# Project Introduction / Background

Clouds play a crucial role in the Earth's climate system by influencing energy balance and weather patterns. However, their complex structures and organization make them difficult to represent accurately in climate models. Understanding different cloud patterns can help scientists improve climate predictions and reduce uncertainties in future climate projections.

In this project, we aim to use satellite images and machine learning techniques to segment and identify different cloud organization patterns. By accurately detecting cloud types and their spatial distribution, this work can contribute to a better understanding of atmospheric processes and support the development of more reliable climate models.

# Chapter 1:

# EDA – Exploratory Data Analysis

## 1.1 Dataset Overview

The dataset used in this project is the Kaggle *Understanding Clouds from Satellite Images* dataset. It consists of satellite images and corresponding cloud segmentation annotations for four cloud categories: **Fish, Flower, Gravel, and Sugar**.

The dataset contains **5,546 unique satellite images** and **22,184 annotation records**, with **5,546 records for each cloud category**. The segmentation annotations are provided in **Run-Length Encoding (RLE)** format through the `EncodedPixels` column of the training CSV file.

The original dataset images have a resolution of **1400 × 2100 pixels**. For this project, the images were processed and resized to **576 × 384 pixels**, corresponding to a NumPy shape of **384 × 576 × 3**.

The dataset is provided through several files:

- `train.csv`: contains the RLE-encoded segmentation masks for each image and cloud-label pair in the training set.
- `train_images.zip`: contains the training satellite images used for model development.
- `test_images.zip`: contains the test satellite images.
- `sample_submission.csv`: provides an example of the required output format for segmentation predictions. This file was not used in the current project.

Not all annotation records contain a segmentation mask. Approximately **53.35% of the annotation records contain valid RLE masks**, while the remaining records have missing `EncodedPixels` values. These missing values indicate that no segmentation mask is provided for that particular cloud category in the corresponding image.

The satellite images were obtained from **NASA Worldview** and were captured by the **TERRA and AQUA** polar-orbiting satellites using the **MODIS** instrument. Because an image may be constructed from two satellite orbits, an uncovered area between successive orbits can appear as a **black band**. This area is not part of the valid image region and must not be considered as a cloud prediction area.

The ground-truth annotations were created through a crowdsourcing process involving **68 scientists**. Each image was labeled by approximately three scientists, and the final ground truth was obtained from the union of their annotations after removing the black band areas.

The dataset therefore provides satellite imagery and pixel-level cloud annotations that can be used to formulate the project as a **multi-label image segmentation task**.

## 1.2 Dataset Structure

The training annotations are stored in a CSV file containing two main columns: `Image_Label` and `EncodedPixels`.

**Table 1.1 Initial dataframe of the training dataset**

| | Image_Label | EncodedPixels |
|---|---|---|
| 0 | 0011165.jpg_Fish | 20056 257 20440 257 20824 257 21208 257 21592 ... |
| 1 | 0011165.jpg_Flower | 102135 1 102245 275 102629 275 103013 275 1033... |
| 2 | 0011165.jpg_Gravel | NaN |
| 3 | 0011165.jpg_Sugar | NaN |
| 4 | 002be4f.jpg_Fish | 17668 241 18051 242 18435 242 18819 242 19203 ... |

The `Image_Label` column contains both the image identifier and the corresponding cloud category. To make the data easier to process, this column was split into two separate columns: `Image_id` and `Cloud_Type`.

**Table 1.2 Dataset structure after separating image identifiers and cloud categories**

| | Image_id | Cloud_Type | EncodedPixels |
|---|---|---|---|
| 0 | 0011165.jpg | Fish | 20056 257 20440 257 20824 257 21208 257 21592 ... |
| 1 | 0011165.jpg | Flower | 102135 1 102245 275 102629 275 103013 275 1033... |
| 2 | 0011165.jpg | Gravel | NaN |
| 3 | 0011165.jpg | Sugar | NaN |
| 4 | 002be4f.jpg | Fish | 17668 241 18051 242 18435 242 18819 242 19203 ... |

For example, an entry such as `0011165.jpg_Fish` is separated into the image identifier `0011165.jpg` and the cloud category `Fish`.

Each image can appear up to four times in the annotation file, once for each cloud category: Fish, Flower, Gravel, and Sugar.

The `EncodedPixels` column contains the segmentation mask information using Run-Length Encoding (RLE). When a cloud of a given category is present and annotated, its pixel locations are stored as an RLE sequence. Missing values in this column indicate that no mask is available for that particular image-category pair.

The RLE-encoded annotations were later decoded and converted into multi-channel segmentation masks for model training. The detailed preprocessing steps are described in Chapter 2.

This structure allows each satellite image to be associated with a corresponding pixel-level target containing the segmentation information for all four cloud categories.

## 1.3 Data Exploration and Visualization

### 1.3.1 Class Distribution

The distribution of available segmentation masks was analyzed for the four cloud categories: Fish, Flower, Gravel, and Sugar. The analysis shows that the number of annotated masks is not exactly equal across the four classes.

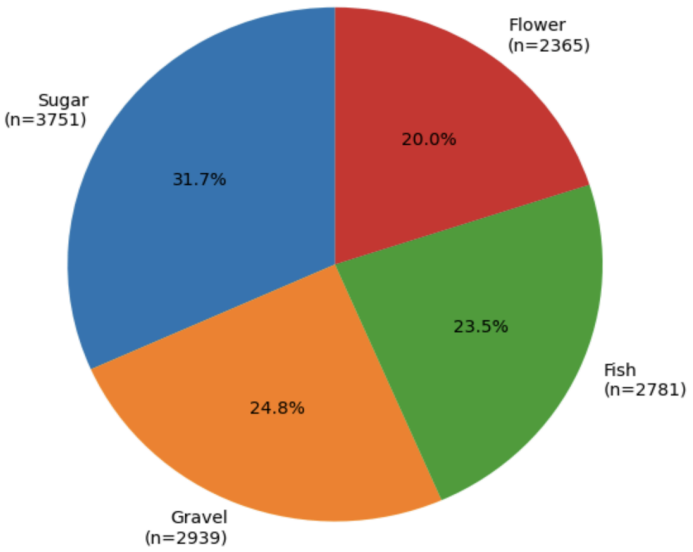



**Figure 1.1: Distribution of available segmentation masks by cloud type**

The pie chart shows the proportion of available masks belonging to each cloud category. Sugar represents the largest proportion, while Flower has the smallest proportion.

Overall, the dataset shows a moderate class imbalance, with Sugar having the highest number of available masks and Flower the lowest. Because the imbalance is not severe, no additional class-balancing technique was applied during preprocessing or model training. In cases of stronger class imbalance, techniques such as class-weighted loss could be used to give greater importance to underrepresented classes. However, for this dataset, the observed imbalance was considered moderate enough to allow the model to learn from the original class distribution without introducing additional weighting.

### 1.3.2 Mask Availability and Cloud Categories per Image

The training dataset contains 22,184 image-category annotation records. Among these records, 11,836 contain segmentation masks, while 10,348 have missing EncodedPixels values. Therefore, 53.35% of the annotation records contain an available segmentation mask, while 46.65% do not.

A missing value indicates that no annotated segmentation mask is available for that particular cloud category in the corresponding image. However, this does not mean that the image contains no clouds, since each image can contain one or more of the four cloud categories.  The number of cloud categories represented by available annotations was also analyzed for each image.

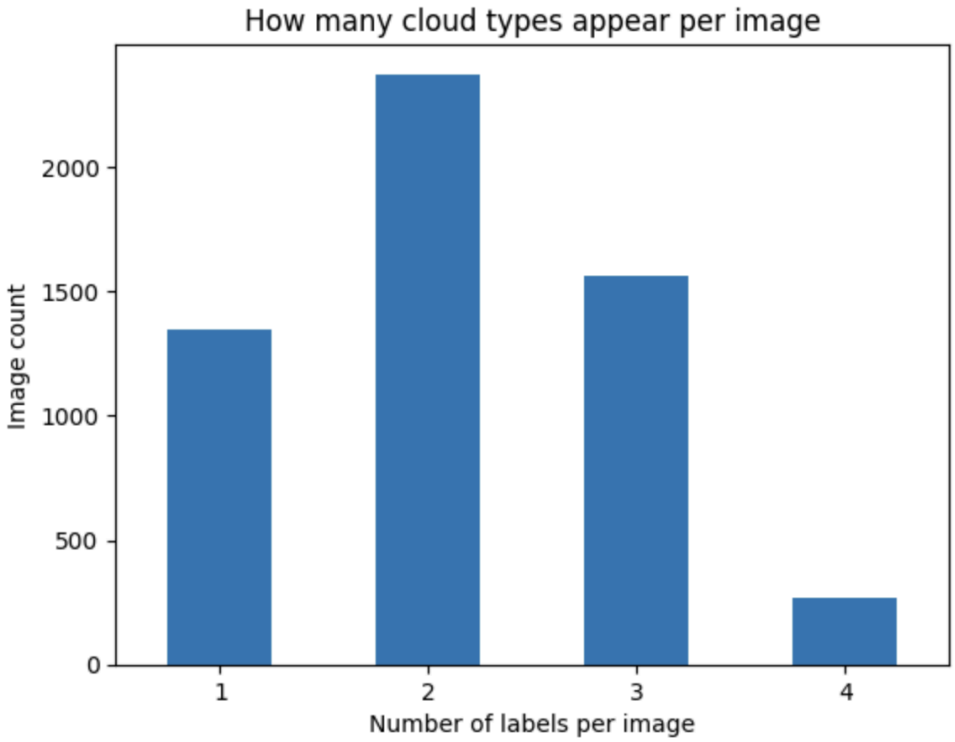



**Figure 1.2: Number of Cloud Types per Image**

The count plot shows the number of cloud types represented by available annotations in each image. The results show that images contain between one and four cloud types. Most images contain two cloud types, followed by three and one cloud type, while only a small number of images contain all four cloud types.

This distribution shows that most satellite images contain multiple cloud types, highlighting the complexity of the segmentation task. The model therefore needs to identify and distinguish several cloud categories within the same image.

### 1.3.3 Sensor Gap Analysis

During the exploration of the satellite images, a black band was observed in some images. This band is caused by an uncovered area between successive satellite orbits and represents a sensor gap rather than a valid image region.

The sensor gap was analyzed to understand the structure of the satellite images and to distinguish the black band from actual cloud regions.

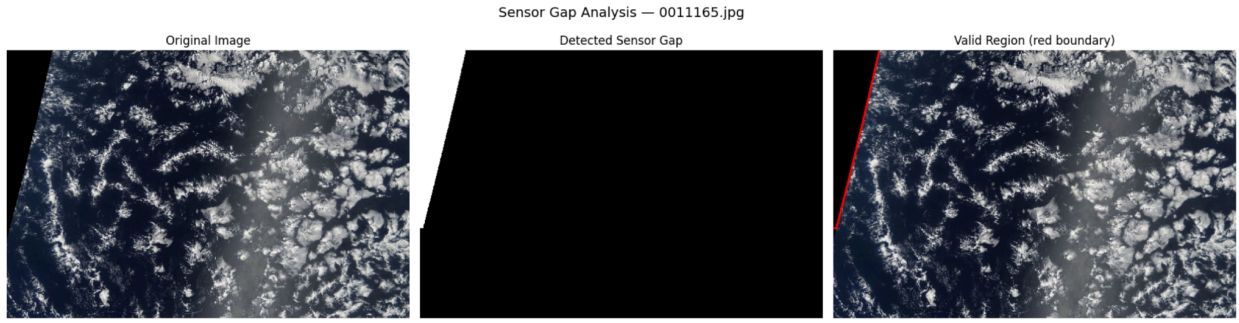



**Figure 1.3: Example of sensor gap detection**

Left: Original satellite image.
Middle: Detected sensor gap (white = gap).
Right: Valid image region with its boundary.

The visualization shows that the black band does not contain valid satellite information and should not be interpreted as a cloud region. The ground truth annotations already exclude these areas.

In the modeling stage, no additional sensor-gap mask was applied during training or prediction. This approach avoids introducing errors from heuristic pixel-based gap detection and keeps the modeling process focused on the provided ground truth annotations.

# Chapter 2

# Data Pre-processing

## 2.1 RLE Decoding

The segmentation masks in the training dataset are provided in Run-Length Encoding (RLE) format through the EncodedPixels column. RLE represents the locations of segmented pixels using pairs of **start positions** and **run lengths**.

To use these annotations for image segmentation, the RLE-encoded values were decoded into binary masks. Each decoded mask has the same spatial dimensions as the processed images, 384 × 576 pixels. Pixels belonging to the corresponding cloud category are assigned a value of 1, while the remaining pixels are assigned a value of 0.

Missing EncodedPixels values were treated as empty masks, since they indicate that no segmentation mask is available for that image-category pair.

The decoded binary masks were then used as the basis for creating the multi-channel ground truth masks described in the next section.

## 2.2 Multi-channel Mask Creation

After decoding the RLE annotations, the binary masks were combined into a single multi-channel mask for each satellite image. Four channels were created, one for each cloud category: Fish, Flower, Gravel, and Sugar.

Each channel represents the segmentation mask of one cloud category, with 1 indicating the presence of the corresponding cloud type and 0 indicating background pixels. Therefore, each processed image is associated with a four-channel ground truth mask with a shape of 384 × 576 × 4.

This representation allows the model to predict the four cloud categories independently for each pixel and provides the ground truth required for multi-label cloud segmentation.

## 2.3 Image and Ground Truth Mask Visualization

To verify the segmentation annotations, a selected satellite image was visualized together with its corresponding ground truth masks. The masks were displayed separately for each cloud category to examine their spatial alignment with the original image.

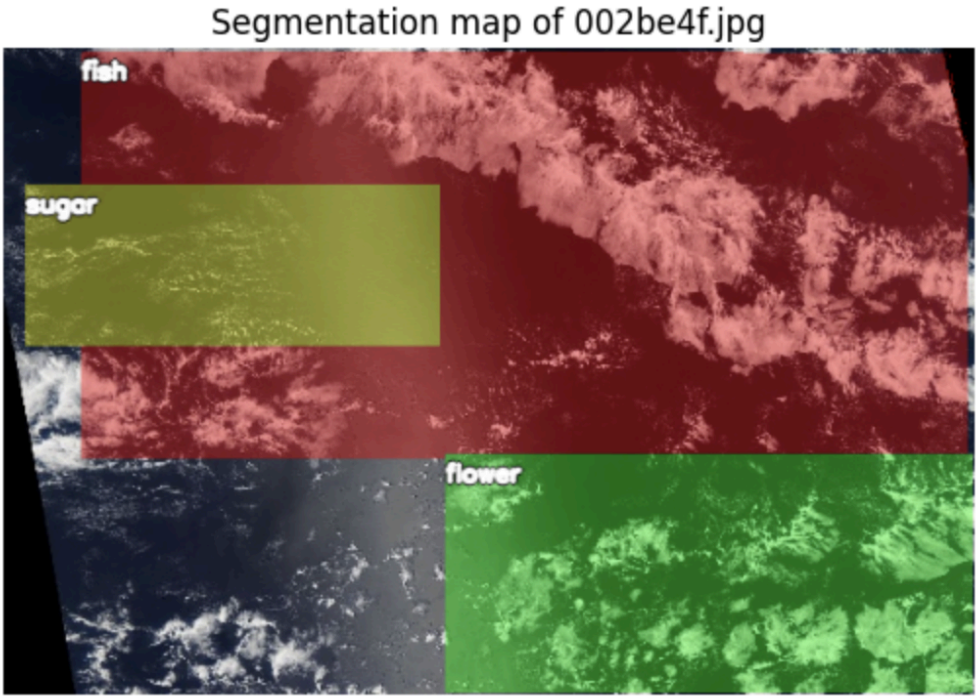



**Figure 2.1: Satellite image 002be4f.jpg with ground truth segmentation masks**

The visualization shows the satellite image 002be4f.jpg together with its available ground truth segmentation masks. The masks illustrate the spatial distribution of the annotated cloud regions and their alignment with the original image.

The black band visible in the lower-left corner represents the sensor gap. It is not part of the valid image region, and the ground truth annotations already exclude this area.

This visualization provides a visual check of the segmentation annotations and confirms their alignment with the satellite image before model training.

## 2.4 Image Resizing

The original satellite images have a resolution of 1400 × 2100 pixels. To reduce the computational cost and make the images suitable for model training, the images were resized to 576 × 384 pixels.

The same spatial dimensions were used for the corresponding ground truth masks to ensure that the images and masks remained correctly aligned. After resizing, each image had a shape of 384 × 576 × 3, while each ground truth mask had a shape of 384 × 576 × 4.

This resizing step reduced the input size while preserving the spatial information required for cloud segmentation and made the dataset more suitable for efficient model training.

## 2.5 Normalization

After resizing, image normalization was applied to prepare the input images for the neural network.
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## 2.6 Data Augmentation

Data augmentation was applied to the training images to increase the diversity of the training data and improve the model's ability to generalize to new satellite images.

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